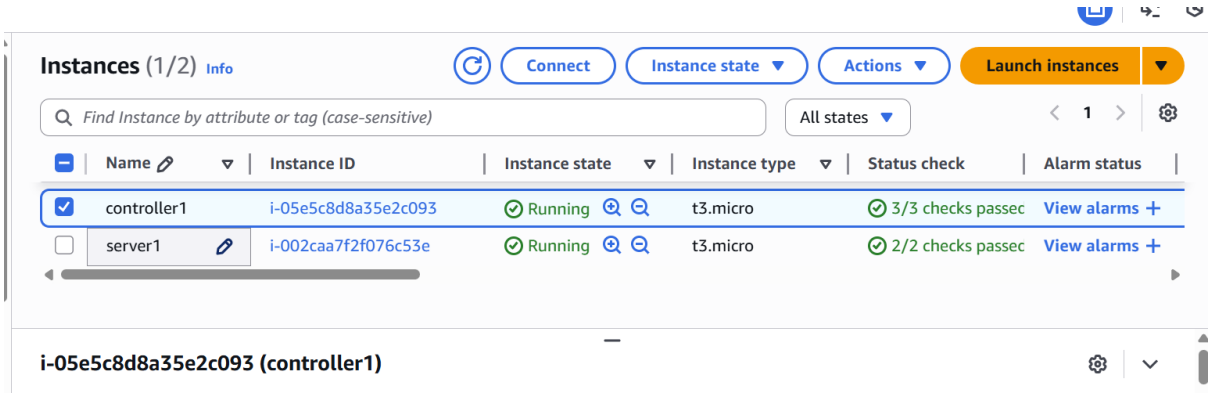
Create 2 instances

A controller and server



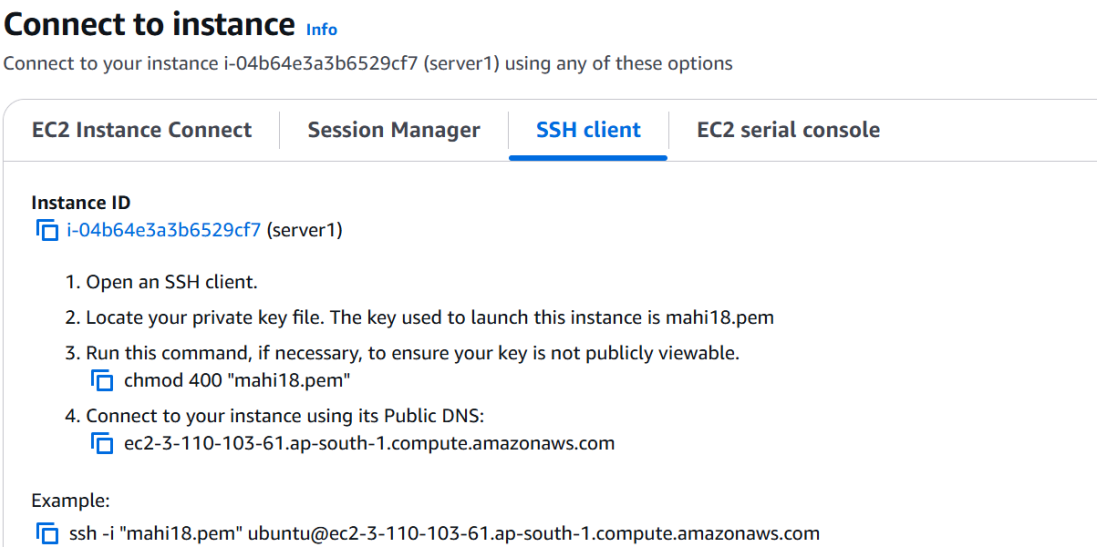
Controller private ipadress : 172.31.1.48

Server ipadress : 172.31.6.55

Connect to server

Ubuntu machine default comes with python 3

But we need pytho 2



```
Admin@DESKTOP-ASGG5M3 MINGW64 ~/Downloads
$ ssh -i "mahi18.pem" ubuntu@ec2-3-110-103-61.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-110-103-61.ap-south-1.compute.amazonaws.com (3.110.103.61)' can't be established.
ED25519 key fingerprint is SHA256:P9CSSuQSFCtJ9ld1aO09ZTMpN2z9dy5/yCYtLrtC17A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-110-103-61.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
```

Update apt repository

Sudo apt-get update

By default u will be having python 3

```
Get:55 http://security.ubuntu.com/ubuntu noble-security/multi
Get:56 http://security.ubuntu.com/ubuntu noble-security/multi
Fetched 40.1 MB in 6s (6385 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-6-55:~$ python3 --version
Python 3.12.3
ubuntu@ip-172-31-6-55:~$ |
```

Install python2

Sudo apt-get install -y python2.7 python-pip

If u find error continue with python 3

```
ubuntu@ip-172-31-5-200:~$ python2 --version
Python 2.7.18
ubuntu@ip-172-31-5-200:~$
```

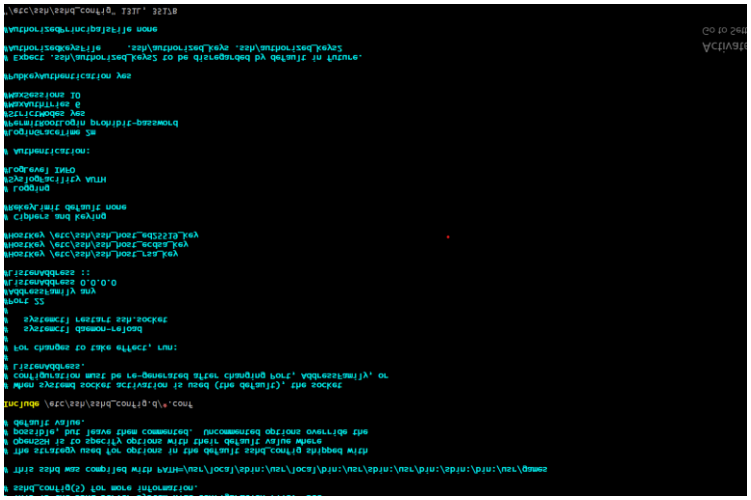
Establish password less connection between controller and managing nodes(server1 and server2)

Sudo passwd ubuntu (give password as ubuntu)

```
ubuntu@ip-172-31-5-200:~$ python2 --version
Python 2.7.18
ubuntu@ip-172-31-5-200:~$ sudo passwd ubuntu
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-5-200:~$
```

Sudo vim /etc/ssh/sshd_config

```
ubuntu@ip-172-31-5-200:~$ sudo vim /etc/ssh/sshd_config
ubuntu@ip-172-31-5-200:~$
```



```
ubuntu@ip-172-31-6-55: ~
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile     .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

-- INSERT --
```

Change password authentication to yes

Save and quit

Restart the service

Sudo service ssh restart

```
ubuntu@ip-172-31-5-200:~$ sudo service ssh restart
ubuntu@ip-172-31-5-200:~$
```

Minimize server

Now connected to controller

Update the repository

Sudo apt-get update

```
Get:34 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:35 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:36 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:37 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:38 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:39 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:40 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backpor
Get:41 http://security.ubuntu.com/ubuntu noble-security/main amd64 P
Get:42 http://security.ubuntu.com/ubuntu noble-security/main Transla
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 C
Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 c
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe amd
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe Tra
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted a
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted T
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted a
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted a
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse a
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse T
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse a
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse a
Fetched 40.1 MB in 6s (6227 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-1-48:~$
```

Install python 2.7 or continue with python 3

Generate the keys

Ssh-keygen

```
ubuntu@ip-172-31-4-180:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:2JP4spYOpoL2k7+1YTRtsiNsocxF7jwE7gkQFzsejqM ubuntu@ip-172-31-4-180
The key's randomart image is:
+---[RSA 3072]----+
|. o.             |
| o .             |
|. +. .           |
|.+.o+  +..       |
|o.o. =o+So       |
|..= O *.=.       |
|E. *.@.o.        |
|...oo Bo+        |
|. .oo+o.         |
+----[SHA256]-----+
ubuntu@ip-172-31-4-180:~$
```

Controller machine open

vim .ssh/id_rsa.pub/ if u find error in key give path of key ie /home/ubuntu/.ssh/id_rsa

or some times u may get /home/ubuntu/.ssh/id_ed25519 this format

```
ubuntu@ip-172-31-1-48: ~
sh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAII6lWOajZ9sm1KD8tTBC6IicWRtvWSXApVqn8gNju+n+ ubuntu@ip-172-31-1-48
```

Connect to server gitbash

vim .ssh/authorized_keys

```
ubuntu@ip-172-31-6-55: ~
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAII6lWOajZ9sm1KD8tTBC6IicWRtvWSXApVqn8gNju+n+ ubuntu@ip-172-31-1-48
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCqRWcE9H1UaqlQvU4Ygkkc6ZTFp3TC+K4FnopjMm9tBHBLJi3IttayMURlrtF1eER7rjgD5WQYmoLtAcuxV5zFnVho8k3YNXxm9nOO3OIn2DgdZv3CZkl+pQM7V37x+jM4hGgN
yeN52oznogoGLxv95Ljp5kn3iT5uWBWldFdgHlHwPLoTn9N6zXetzgi33b2xtKlHrum/1ZqG4YsEi7I52Fae0TsZ4D3x+NVYkXPHWHBesoGnV++sRp3tplSt0lkkXN3l4PUAMtNKCaS2ETibx9gmlT++W4u55ocpA+PcJudyMzz4P
VTfkOKOwygc1RLTr4y5whhygDPg2Egy9F4ZX bhavani
~
~
```

Connect to controller machine

ssh -v ubuntu@server1 ip address

```
ubuntu@ip-172-31-1-48:~$ vim /home/ubuntu/.ssh/id_ed25519
ubuntu@ip-172-31-1-48:~$ vim /home/ubuntu/.ssh/id_ed25519.p
ubuntu@ip-172-31-1-48:~$ vim /home/ubuntu/.ssh/id_ed25519.p
ubuntu@ip-172-31-1-48:~$ ssh -v ubuntu@172.31.6.55
```

```
Last login: Thu Feb 19 12:50:40 2026 from 157.66.152.5
ubuntu@ip-172-31-6-55:~$ |
```

Here u cam see ipadress of server

Enter exit to move into controller

```
Last login: Thu Feb 19 12:50:40 2026 from 157.66.152.5
ubuntu@ip-172-31-6-55:~$ exit
logout
debug1: client_input_channel_req: channel 0 rtype exit-status reply 0
debug1: client_input_channel_req: channel 0 rtype eow@openssh.com reply 0
debug1: channel 0: free: client-session, nchannels 1
Connection to 172.31.6.55 closed.
Transferred: sent 7864, received 9148 bytes, in 155.5 seconds
Bytes per second: sent 50.6, received 58.8
debug1: Exit status 0
ubuntu@ip-172-31-1-48:~$
```

Connect to controller

Install the ansible

sudo apt-get install software-properties-common

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install software-properties-common
Reading package lists... Done
```

sudo apt-add-repository ppa:ansible/ansible

```
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
ubuntu@ip-172-31-4-180:~$ sudo apt-add-repository ppa:ansible/ansible
Repository: 'deb https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/ jam
```

sudo apt-get update

sudo apt-get install -y ansible

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Redirect to inventory files

cd /etc/ansible

$ll

```
drwxr-xr-x   3 root root 4096 Feb 19 13:14 ./
drwxr-xr-x 109 root root 4096 Feb 19 13:14 ../
-rw-r--r--   1 root root  614 Jan 30 18:08 ansible.cfg
-rw-r--r--   1 root root 1175 Jan 30 18:08 hosts
drwxr-xr-x   2 root root 4096 Jan 30 18:08 roles/
ubuntu@ip-172-31-1-48:/etc/ansible$ |
```

sudo vim hosts

```
ubuntu@ip-172-31-1-48: /etc/ansible
# This is the default ansible 'hosts' file.
#
# It should live in /etc/ansible/hosts
#
#   - Comments begin with the '#' character
#   - Blank lines are ignored
#   - Groups of hosts are delimited by [header] elements
#   - You can enter hostnames or ip addresses
#   - A hostname/ip can be a member of multiple groups

# Ex 1: Ungrouped hosts, specify before any group headers:

## green.example.com
## blue.example.com
## 192.168.100.1
## 192.168.100.10

# Ex 2: A collection of hosts belonging to the 'webservers' group:

## [webservers]
## alpha.example.org
## beta.example.org
## 192.168.1.100
## 192.168.1.110

# If you have multiple hosts following a pattern, you can specify
# them like this:

## www[001:006].example.com

# You can also use ranges for multiple hosts:

## db-[99:101]-node.example.com

# Ex 3: A collection of database servers in the 'dbservers' group:

## [dbservers]
##
## db01.intranet.mydomain.net
## db02.intranet.mydomain.net
## 10.25.1.56
## 10.25.1.57


# Ex4: Multiple hosts arranged into groups such as 'Debian' and 'openSUSE':

## [Debian]
## alpha.example.org
## beta.example.org

## [openSUSE]
## green.example.com
## blue.example.com
```

Paste private ipadress of server

```
ubuntu@ip-172-31-1-48: /etc/ansible
172.31.6.55
# This is the default ansible 'hosts' file.
#
# It should live in /etc/ansible/hosts
#
#   - Comments begin with the '#' character
#   - Blank lines are ignored
#   - Groups of hosts are delimited by [header] elements
#   - You can enter hostnames or ip addresses
#   - A hostname/ip can be a member of multiple groups

# Ex 1: Ungrouped hosts, specify before any group headers:

## green.example.com
```

Adhoc commands: it works based on modules

Command module:

Create files in server1 and 2 from controller

ansible all -i /etc/ansible/hosts -m command -a 'touch f1 f2 f3'

```
-rw-r--r--   1 root root 1175 Jan 30 18:08 hosts
drwxr-xr-x   2 root root 4096 Jan 30 18:08 roles/
ubuntu@ip-172-31-1-48:/etc/ansible$ sudo vim hosts
ubuntu@ip-172-31-1-48:/etc/ansible$ sudo vim hosts
ubuntu@ip-172-31-1-48:/etc/ansible$ ansible all -i /etc/ansible/hosts -m command -a 'touch f1 f2 f3'
[WARNING]: Host '172.31.6.55' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a diffe
rent interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
172.31.6.55 | CHANGED | rc=0 >>

ubuntu@ip-172-31-1-48:/etc/ansible$
```

To check files are created or not

Ssh give your server private ipadress

$ll

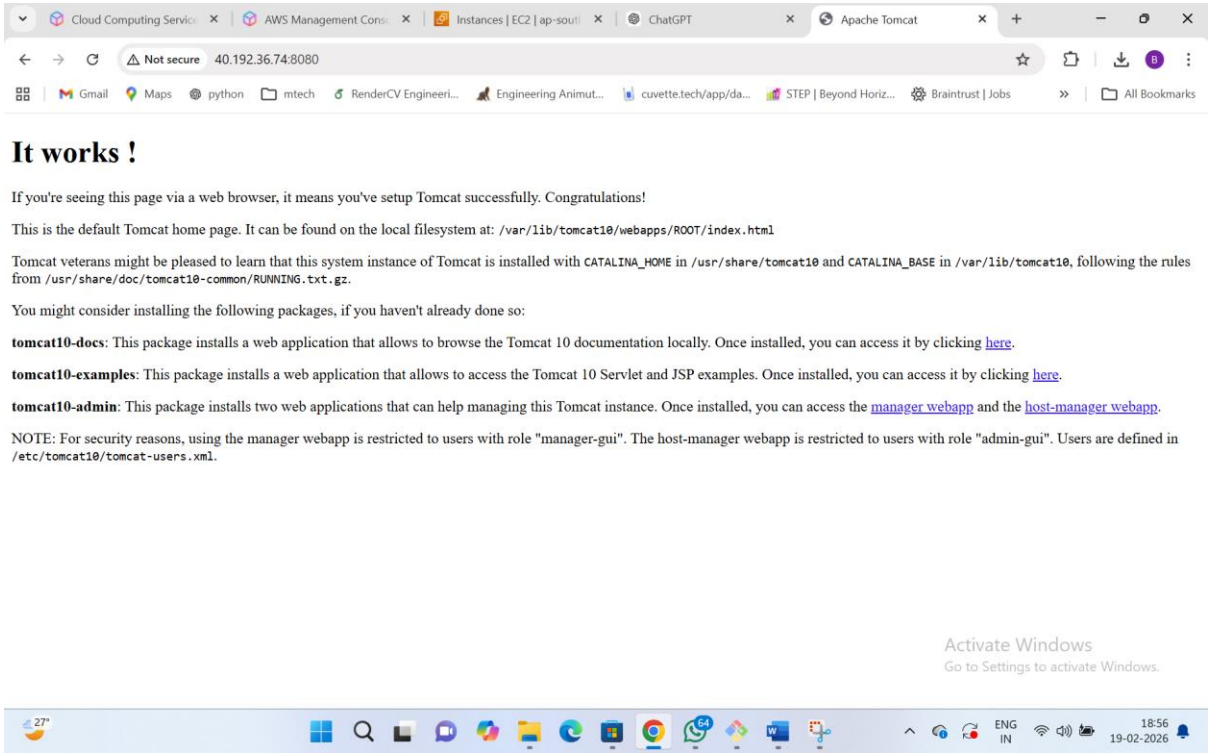
```
ubuntu@ip-172-31-6-55:~$ ll
total 40
drwxr-x--- 5 ubuntu ubuntu 4096 Feb 19 13:18 ./
drwxr-xr-x 3 root   root   4096 Feb 19 12:47 ../
drwx------ 3 ubuntu ubuntu 4096 Feb 19 13:18 .ansible/
-rw------- 1 ubuntu ubuntu    5 Feb 19 13:10 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx------ 2 ubuntu ubuntu 4096 Feb 19 12:50 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx------ 2 ubuntu ubuntu 4096 Feb 19 13:06 .ssh/
-rw-r--r-- 1 ubuntu ubuntu    0 Feb 19 12:51 .sudo_as_admin_successful
-rw------- 1 ubuntu ubuntu  802 Feb 19 13:06 .viminfo
-rw-rw-r-- 1 ubuntu ubuntu    0 Feb 19 13:18 f1
-rw-rw-r-- 1 ubuntu ubuntu    0 Feb 19 13:18 f2
-rw-rw-r-- 1 ubuntu ubuntu    0 Feb 19 13:18 f3
ubuntu@ip-172-31-6-55:~$
```

Install tomcat

ansible all -m apt -a 'name=tomcat9 state=present update_cache=yes' -b (use tomcat9)

in some cases go for tomcat 10 if u get any error in tomcat 9

```
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=tomcat8 state=present update_cache=yes' -b
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.7.139 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "msg": "No package matching 'tomcat8' is available"
}
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "msg": "No package matching 'tomcat8' is available"
}
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=tomcat9 state=present update_cache=yes' -b
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1737560819,
    "cache_updated": true,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following packages were automatically installed and ar
quired:\n  git-man liberror-perl\nUse 'sudo apt autoremove' to remove them.\nThe following additional packages will be installed:\n  alsa-topology-conf alsa-u
tificates-java default-jre-headless\n  java-common libapr1 libasound2 libasound2-data libavahi-client3\n  libavahi-common-data libavahi-common3 libcups2 libec
-java\n  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1\n  libtomcat9-java openjdk-11-jre-headless tomcat9-common\nSuggested packages:\n
basound2-plugins alsa-utils cups-common liblcms2-utils pcscd\n  libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho\n  fonts-wqy-microhei
enhei fonts-indic tomcat9-admin tomcat9-docs\n  tomcat9-examples tomcat9-user\nThe following NEW packages will be installed:\n  alsa-topology-conf alsa-ucm-co
ates-java default-jre-headless\n  java-common libapr1 libasound2 libasound2-data libavahi-client3\n  libavahi-common-data libavahi-common3 libcups2 libeclipse
\n  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1\n  libtomcat9-java openjdk-11-jre-headless tomcat9 tomcat9-common\n0 upgraded, 22 newly
```

Give server public ipadress and port no :8080

Ipadress:8080